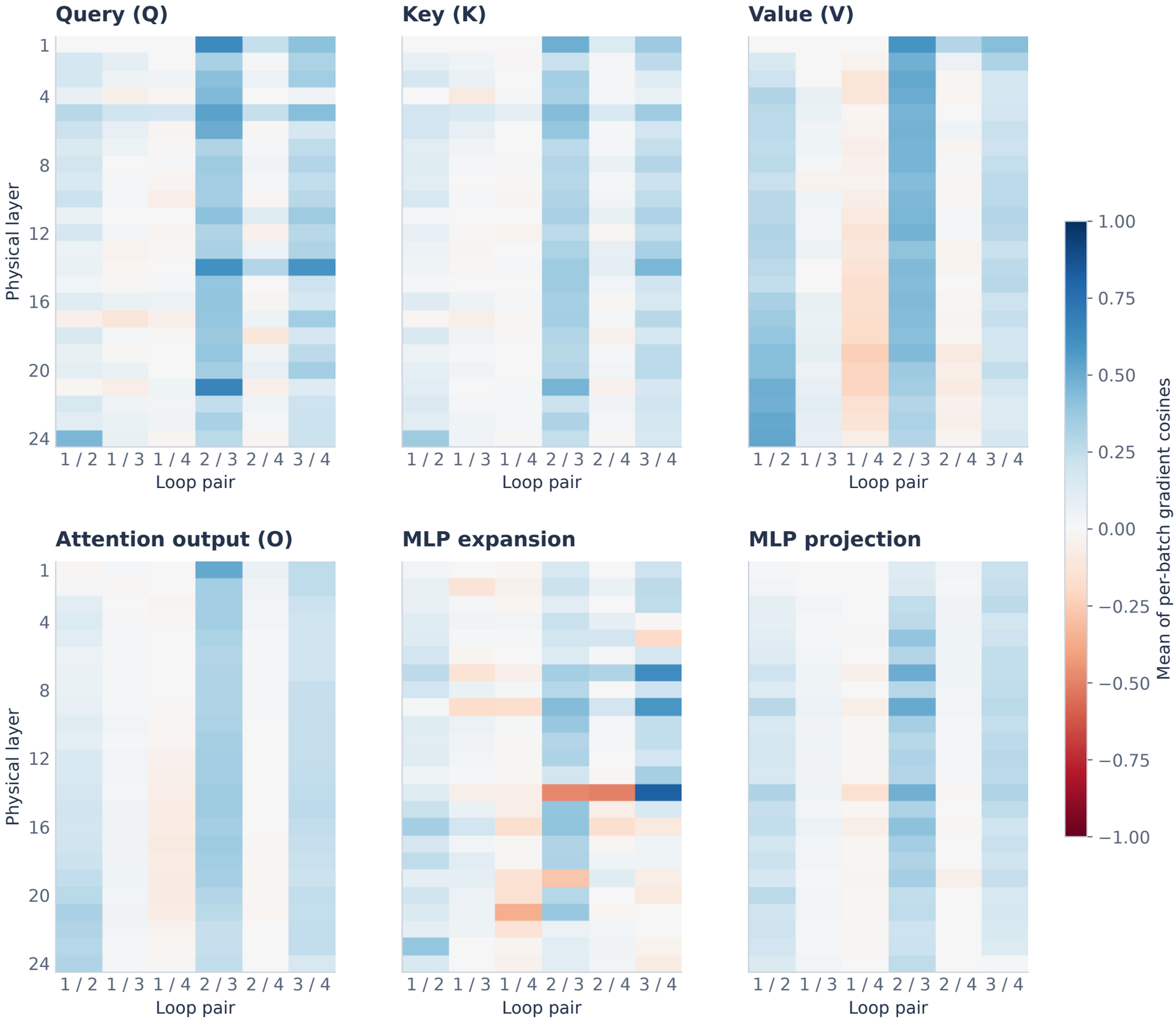
Do different loops request the same parameter update?

Cosine within each mini-batch, then average across 32 batches.



Red = opposing; blue = aligned. Not the cosine of the mean gradients.
Final checkpoint; 32 batches. Disagreement alone does not establish sharing harm.